

0 7 . 1

Describe binary fission in bacteria.

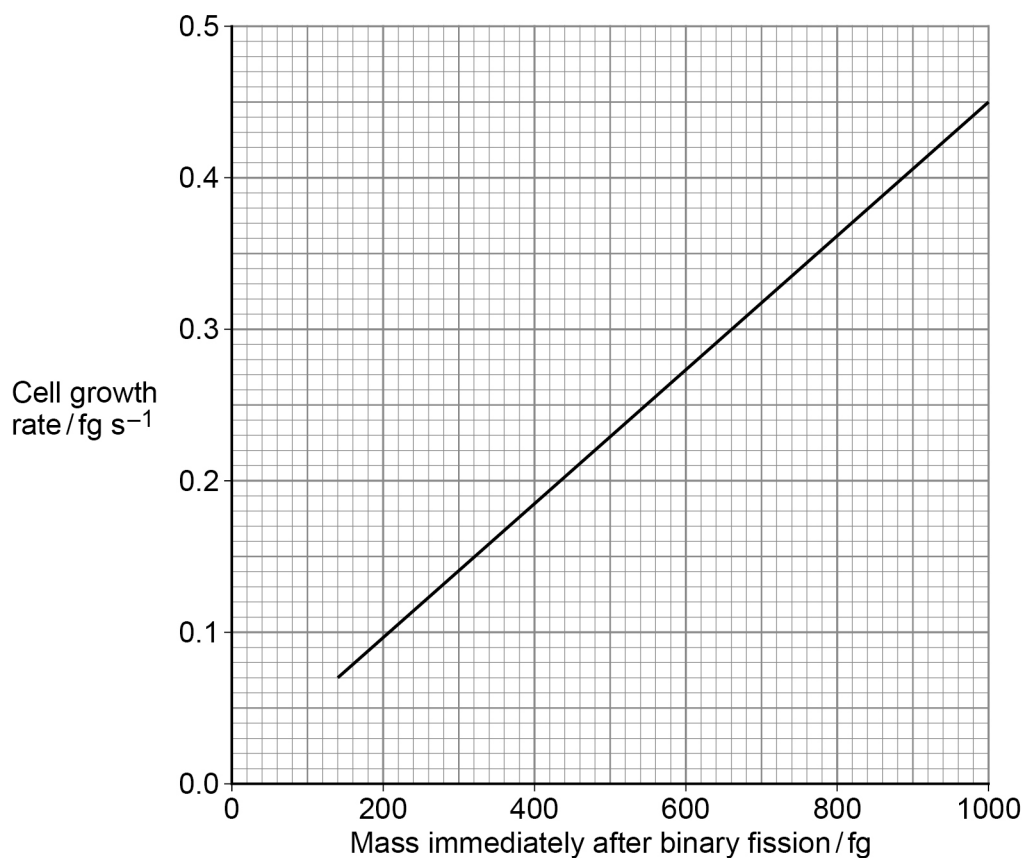
[3 marks]

Question 7 continues on the next page**Turn over ►**

The cell growth rate of the bacterium *Bacillus subtilis* is proportional to its mass immediately after binary fission.

Figure 7 shows this relationship.

Figure 7



0 7 . 2 The mass of the bacterial cells was measured in femtograms (fg).

1 fg (femtogram) = 1×10^{-15} g

Place a tick (✓) in the box next to the number that is equal to 680 fg

[1 mark]

0.000 000 000 006 8 g

☐

6.8×10^{-13} g

☐

6.8×10^{-15} g

☐

6.8×10^{-17} g

☐


A scientist determined the growth rate of a *B. subtilis* cell by measuring its mass for 5 minutes.

In those 5 minutes, the cell's mass increased by 90 fg

0 7 . 3

Use this information and **Figure 7** to determine the mass of this cell immediately after binary fission.

Show your working.

[2 marks]

Answer _____ fg

0 7 . 4

Suggest and explain how **two** environmental variables could be changed to increase the growth rate of these cells.

[4 marks]

Suggestion 1 _____

Explanation _____

Suggestion 2 _____

Explanation _____

10

Turn over ►

